



## **SITYOG INSTITUTE OF TECHNOLOGY**

Growth Centre, Jasoiya More, Aurangabad, Bihar – 824101

### **DEPARTMENT OF COMPUTER APPLICATION**

Report on

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## **STUDENT TRAINING MANAGEMENT SYSTEM**

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This report is submitted in partial fulfilment of the requirements for  
the award of the degree

### **Bachelor of Computer Application**

**Submitted by:**

**Abhishek Kumar**

23303106037

**Anand Kumar Pal**

23303106005

**Chirag Singh**

23303106010

**Mohit Raj**

23303106022

**Om Prakash Kumar**

23303106053

**Submitted to:**

**Mr. S.K Ojha**

Assistant Professor

Department of Computer Applications

SITYOG Institute of Technology

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**DEPARTMENT OF COMPUTER APPLICATION**

**SITYOG INSTITUTE OF TECHNOLOGY, AURANGABAD, BIHAR – 824101**

**JULY 2025**

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## **DECLARATION CERTIFICATE**

This is to certify the work presented in the Project Report in "**Student Training Management System**", in partial fulfilment of the requirement for the award of Degree of Bachelor of Computer Application of SITYOG Institute of Technology, Aurangabad, is an authentic work carried out by us under the supervision of **Mr. S.K Ojha**, Assistant Professor, Department of Computer Applications.

To the best of our knowledge, the content of this report does not form a basis for the award of any previous Degree to anyone else.

Date: .....

**Abhishek Kumar**

23303106037

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|                         |             |
|-------------------------|-------------|
| <b>Anand Kumar Pal</b>  | 23303106005 |
| <b>Chirag Singh</b>     | 23303106010 |
| <b>Mohit Raj</b>        | 23303106022 |
| <b>Om Prakash Kumar</b> | 23303106053 |

**Branch:** Bachelor of Computer Application   **Semester:** 6th   **Year:** 2025–26

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Project Supervisor

## **CERTIFICATE OF APPROVAL**

The foregoing Project REPORT on "**Student Training Management System**" is hereby approved as a creditable study of Project report and has been presented in satisfactory manner to warrant its acceptance as prerequisite to the degree for which it has been submitted.

It is understood that by this undersigned do not necessarily endorse any conclusion drawn or opinion expressed therein, but approve the report for the purpose for which it is submitted.

|                         |             |
|-------------------------|-------------|
| <b>Abhishek Kumar</b>   | 23303106037 |
| <b>Anand Kumar Pal</b>  | 23303106005 |
| <b>Chirag Singh</b>     | 23303106010 |
| <b>Mohit Raj</b>        | 23303106022 |
| <b>Om Prakash Kumar</b> | 23303106053 |

**Branch:** Bachelor of Computer Application

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**Year:** 2025–26

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Internal Examiner

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External Examiner

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Head of Department

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|                         |             |
|-------------------------|-------------|
| <b>Abhishek Kumar</b>   | 23303106037 |
| <b>Anand Kumar Pal</b>  | 23303106005 |
| <b>Chirag Singh</b>     | 23303106010 |
| <b>Mohit Raj</b>        | 23303106022 |
| <b>Om Prakash Kumar</b> | 23303106053 |

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## Chapter 1

# Introduction

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### **1.1 Background**

The rapid advancement of information technology has transformed the way educational institutions manage their administrative and academic functions. Traditionally, student records, training programmes, and academic results were maintained manually through paper registers and spreadsheets — methods that are time-consuming, error-prone, and difficult to retrieve quickly.

In the context of technical and professional education, managing training and internship records is particularly challenging. Students are placed with various companies as part of their curriculum, and keeping track of each student's training status, project, company, and performance requires a well-organised system.

The **Student Training Management System** is developed to address these challenges by providing a centralised, web-based platform for managing student personal details, company information, training records, and academic results.

### **1.2 Problem Statement**

The following problems were identified at educational institutions like Sityog Institute of Technology:

- Manual record-keeping is slow and error-prone; data retrieval is tedious.
- No centralised system exists to track which students are training with which companies.
- Academic results are stored separately without link to training records.
- Generating consolidated reports is laborious and time-consuming.
- No secure authentication to prevent unauthorised access to student data.

### **1.3 Objectives**

- **Develop a web-based system** for managing student details, training records, company information, and academic results.
- **Implement secure authentication** so only authorised administrators can access and modify data.
- **Provide real-time search and filtering** for quick access to records by name, roll number, or other parameters.
- **Maintain training and internship records** including company, project, duration, and status.
- **Track academic results** across semesters and generate result summaries.
- **Generate comprehensive reports** covering all modules in a consolidated view.
- **Build a modern, responsive UI** that is intuitive for administrative staff.

### **1.4 Scope of the Project**

#### **In Scope:**

- Management of student personal and academic information.
- Management of company and placement records.
- Tracking of student training and internship programmes.
- Recording and viewing of semester-wise academic results.
- Consolidated report generation for all modules.
- Secure admin login and JWT session management.
- Individual student profile view with complete history.

#### **Out of Scope:**

- Student-facing portal (students cannot log in in this version).
- Fee or payment management.
- Attendance management.
- Email or SMS notification system.
- Multi-campus or multi-department support.

## Literature Review

---

### 2.1 Existing Systems

#### 2.1.1 Traditional Manual Systems

Many small and medium-sized educational institutions in India still rely on paper registers and files. While simple, these methods require significant storage space, make retrieval tedious, and are highly susceptible to data loss due to damage or misplacement.

#### 2.1.2 Microsoft Excel-Based Systems

Spreadsheets offer an improvement over paper-based records but lack multi-user access, role-based security, and the ability to generate dynamic cross-module reports. They also have no validation and are prone to accidental modification.

#### 2.1.3 Enterprise ERP Systems

Large universities use enterprise ERP systems such as SAP or Oracle. While comprehensive, these are prohibitively expensive, require dedicated IT infrastructure, and are far too complex for small institutes with limited technical staff.

#### 2.1.4 Desktop Applications (VB.NET / Access)

Desktop-based applications built with Visual Basic .NET and Microsoft Access are widely used in college projects. However, they are Windows-only, require local installation, and cannot be accessed remotely — making them unsuitable for modern institutional requirements.

### 2.2 Proposed System

The proposed system overcomes the above limitations through modern web technologies:

- **Web-based:** Accessible from any browser without installation.
- **Cloud Database:** Neon PostgreSQL ensures data availability, backups, and scalability.
- **Secure Auth:** NextAuth v5 with JWT and bcrypt password hashing.
- **Modern UI:** shadcn/ui with Tailwind CSS — clean, responsive, professional.
- **Real-time Search:** Instant filtering across all modules.
- **REST API:** Clean separation of frontend and backend, easy to extend.

### **2.3 Comparison of Existing and Proposed System**

*Table 2.1: Comparison of Existing and Proposed Systems*

| Feature               | Manual   | Excel         | VB.NET App   | Proposed System |
|-----------------------|----------|---------------|--------------|-----------------|
| Platform              | Paper    | Windows       | Windows Only | Any Browser     |
| Multi-user Access     | No       | Limited       | No           | Yes             |
| Data Security         | None     | Password only | Basic        | JWT + bcrypt    |
| Search & Filter       | Manual   | Basic         | Limited      | Real-time       |
| Report Generation     | Manual   | Manual        | Basic        | Automated       |
| Cloud Storage         | No       | No            | No           | Yes (Neon DB)   |
| Remote Access         | No       | No            | No           | Yes             |
| Installation Required | No       | Yes           | Yes          | No              |
| Training Records      | Register | Sheet         | Basic        | Full CRUD       |
| Result Management     | Register | Sheet         | Basic        | Full CRUD       |
| Cost                  | Low      | Low           | Medium       | Low (free tier) |

## System Analysis and Requirements

---

### 3.1 Feasibility Study

#### 3.1.1 Technical Feasibility

The system uses Next.js 16, TypeScript, Prisma ORM, and Neon PostgreSQL — all open-source, well-documented, and widely supported technologies. Development requires only a computer with Node.js installed. The cloud database provides a free tier sufficient for institutional use. The project is technically feasible.

#### 3.1.2 Economic Feasibility

All technologies are free and open-source. Neon PostgreSQL offers a generous free tier with no licensing fees. Development was carried out by the student team as an academic project. The project is economically feasible at near-zero cost.

#### 3.1.3 Operational Feasibility

The system's intuitive UI requires minimal training for administrative staff. Being web-based, it can be accessed from any device without software installation. The project is operationally feasible.

### 3.2 Functional Requirements

*Table 3.1: Functional Requirements*

| Req. ID | Module    | Requirement                                             |
|---------|-----------|---------------------------------------------------------|
| FR-01   | Auth      | Admin must log in with username and password            |
| FR-02   | Auth      | Unauthenticated users must be redirected to login       |
| FR-03   | Students  | Admin can add, edit, delete, and view student records   |
| FR-04   | Students  | Search students by name or roll number                  |
| FR-05   | Students  | View individual student profile with training & results |
| FR-06   | Companies | Admin can add, edit, delete, and view company records   |
| FR-07   | Training  | Admin can add, edit, delete, and view training records  |
| FR-08   | Training  | Training status: Ongoing or Completed                   |

| Req. ID | Module  | Requirement                                          |
|---------|---------|------------------------------------------------------|
| FR-09   | Results | Admin can add, edit, delete, and view result records |
| FR-10   | Reports | Consolidated report view for all modules             |

### **3.3 Non-Functional Requirements**

*Table 3.2: Non-Functional Requirements*

| Req. ID | Category        | Requirement                                      |
|---------|-----------------|--------------------------------------------------|
| NFR-01  | Security        | All routes protected by JWT authentication       |
| NFR-02  | Security        | Passwords stored as bcrypt hashes only           |
| NFR-03  | Performance     | Page load under 3 seconds on standard connection |
| NFR-04  | Usability       | Clean, consistent UI across all modules          |
| NFR-05  | Reliability     | All API errors return meaningful messages        |
| NFR-06  | Scalability     | Database capable of handling growth in records   |
| NFR-07  | Compatibility   | Works on Chrome, Firefox, Edge, and Safari       |
| NFR-08  | Maintainability | Modular, consistent code conventions             |

### **3.4 Use Case Description**

*Table 3.3: Use Case Table – Admin*

| Use Case       | Actor | Description                                    | Outcome                            |
|----------------|-------|------------------------------------------------|------------------------------------|
| Login          | Admin | Enter credentials to access system             | Authenticated session created      |
| Logout         | Admin | End the session                                | Session cleared, redirect to login |
| Add Student    | Admin | Fill student form and submit                   | Student created in database        |
| Edit Student   | Admin | Modify existing student details                | Student record updated             |
| Delete Student | Admin | Remove student from system                     | Record permanently deleted         |
| Search         | Admin | Type in search box to filter records           | Filtered results in real time      |
| View Profile   | Admin | Click eye icon on student row                  | Individual profile page shown      |
| Add Training   | Admin | Select student, company, fill training details | Training record created            |
| Add Result     | Admin | Select student, enter result details           | Result record created              |

| Use Case     | Actor | Description              | Outcome                        |
|--------------|-------|--------------------------|--------------------------------|
| View Reports | Admin | Navigate to reports page | All module data shown together |



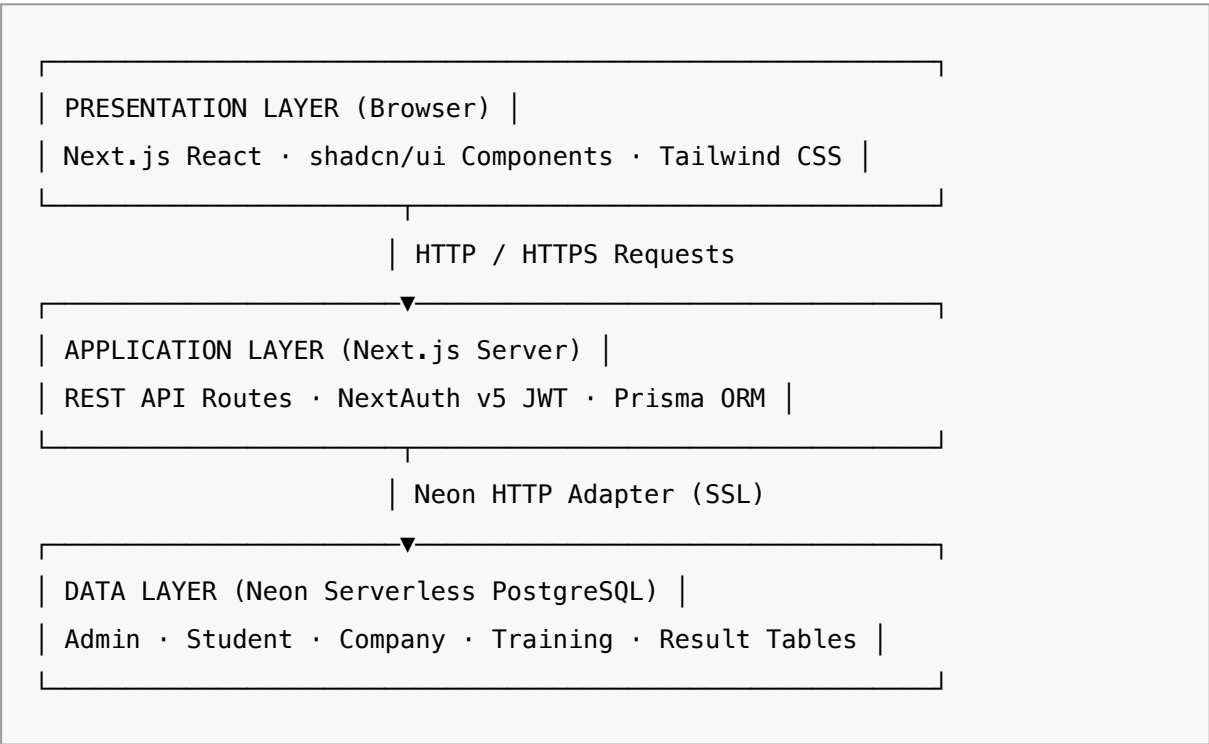
## System Design

### 4.1 System Architecture

The system follows a **three-tier architecture** — Presentation, Application, and Data layers.

- **Presentation Layer:** React components (Next.js App Router) with shadcn/ui and Tailwind CSS handle all user interactions.
- **Application Layer:** Next.js API Routes implement the REST API. NextAuth v5 manages authentication. The proxy file enforces route protection at the edge.
- **Data Layer:** PostgreSQL on Neon DB, accessed via Prisma ORM with the Neon HTTP adapter for serverless-compatible queries.

*Figure 4.1: System Architecture – Three-Tier Web Application*



### 4.2 Database Design

*Table 4.1: Student Table Schema*

| Column | Type    | Constraints        | Description         |
|--------|---------|--------------------|---------------------|
| id     | INT     | PK, AUTO INCREMENT | Unique identifier   |
| rollNo | VARCHAR | UNIQUE, NOT NULL   | Student roll number |

| Column    | Type      | Constraints   | Description           |
|-----------|-----------|---------------|-----------------------|
| name      | VARCHAR   | NOT NULL      | Full name             |
| email     | VARCHAR   | NOT NULL      | Email address         |
| phone     | VARCHAR   | NOT NULL      | Contact number        |
| address   | VARCHAR   | NOT NULL      | Home address          |
| course    | VARCHAR   | NOT NULL      | Enrolled course       |
| year      | INT       | NOT NULL      | Current year (1/2/3)  |
| createdAt | TIMESTAMP | DEFAULT NOW() | Creation timestamp    |
| updatedAt | TIMESTAMP | AUTO UPDATE   | Last update timestamp |

**Table 4.2: Company Table Schema**

| Column        | Type    | Constraints        | Description       |
|---------------|---------|--------------------|-------------------|
| id            | INT     | PK, AUTO INCREMENT | Unique identifier |
| name          | VARCHAR | UNIQUE, NOT NULL   | Company name      |
| industry      | VARCHAR | NOT NULL           | Industry sector   |
| location      | VARCHAR | NOT NULL           | City / location   |
| contactPerson | VARCHAR | NOT NULL           | HR contact name   |
| email         | VARCHAR | NOT NULL           | Contact email     |
| phone         | VARCHAR | NOT NULL           | Contact phone     |

**Table 4.3: Training Table Schema**

| Column      | Type      | Constraints               |
|-------------|-----------|---------------------------|
| id          | INT       | PK, AUTO INCREMENT        |
| studentId   | INT       | FK → Student.id, NOT NULL |
| companyId   | INT       | FK → Company.id, NOT NULL |
| projectName | VARCHAR   | NOT NULL                  |
| startDate   | TIMESTAMP | NOT NULL                  |
| endDate     | TIMESTAMP | NOT NULL                  |
| status      | VARCHAR   | DEFAULT 'Ongoing'         |
| description | TEXT      | NOT NULL                  |

**Table 4.4: Result Table Schema**

| Column    | Type    | Constraints               |
|-----------|---------|---------------------------|
| id        | INT     | PK, AUTO INCREMENT        |
| studentId | INT     | FK → Student.id, NOT NULL |
| subject   | VARCHAR | NOT NULL                  |
| marks     | INT     | NOT NULL                  |

| Column   | Type    | Constraints |
|----------|---------|-------------|
| maxMarks | INT     | NOT NULL    |
| grade    | VARCHAR | NOT NULL    |
| semester | INT     | NOT NULL    |
| year     | INT     | NOT NULL    |

### **4.3 Module Design**

Each module has a dedicated page under `/dashboard/` and corresponding REST API routes under `/api/`:

- **Authentication:** `/login` page · `/api/auth` (NextAuth handler)
- **Dashboard:** `/dashboard` · `/api/dashboard` (summary counts)
- **Students:** `/dashboard/students` · `/api/students` + `/api/students/[id]`
- **Companies:** `/dashboard/companies` · `/api/companies` + `/api/companies/[id]`
- **Training:** `/dashboard/training` · `/api/training` + `/api/training/[id]`
- **Results:** `/dashboard/results` · `/api/results` + `/api/results/[id]`
- **Reports:** `/dashboard/reports` · reads from all existing API endpoints

# Implementation

## 5.1 Technology Stack

Table 5.1: Technology Stack Details

| Layer            | Technology            | Version    | Purpose                                                   |
|------------------|-----------------------|------------|-----------------------------------------------------------|
| Framework        | Next.js               | 16.2.10    | Full-stack React framework with App Router and API Routes |
| Language         | TypeScript            | 5.x        | Statically typed JavaScript for safer, maintainable code  |
| Styling          | Tailwind CSS          | 4.x        | Utility-first CSS for rapid, responsive UI development    |
| UI Components    | shadcn/ui             | Latest     | Accessible, composable component library                  |
| Icons            | Lucide React          | Latest     | SVG icon set                                              |
| ORM              | Prisma                | 7.8.0      | Type-safe database access layer                           |
| DB Adapter       | @prisma/adapters-neon | 7.8.0      | Serverless-compatible Neon adapter for Prisma             |
| Database         | Neon PostgreSQL       | 15         | Serverless cloud PostgreSQL database                      |
| Authentication   | NextAuth v5           | 5.0.0-beta | JWT session management with Credentials provider          |
| Password Hashing | bcryptjs              | 3.x        | Secure password hashing                                   |

## 5.2 Module Implementation with Screenshots

### 5.2.1 Login Page

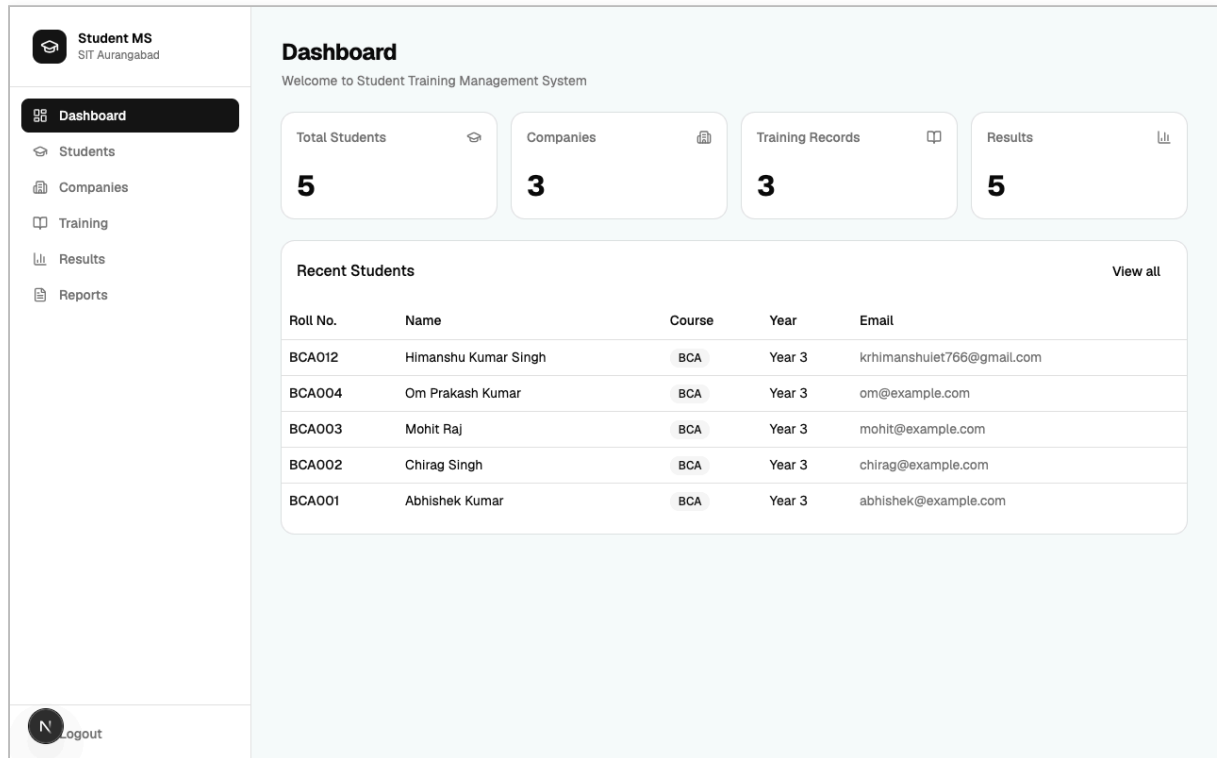
The login page is the entry point of the application presenting a card-based form. On successful authentication, a JWT session is created and the admin is redirected to the dashboard. Invalid credentials display an error message.



Figure 5.1: Login Page

### 5.2.2 Dashboard

The dashboard displays four summary cards (total students, companies, training records, and results) and a table of recently added student records. The sidebar provides navigation to all modules.



*Figure 5.2: Dashboard – Overview and Statistics*

### 5.2.3 Student Module

The student module provides a searchable table with roll number, name, email, phone, course, and year. A right-side drawer panel handles add/edit forms without page navigation. An eye icon navigates to the student's full profile page.

Student MS

SIT Aurangabad

Dashboard

Students

Companies

Training

Results

Reports

N

Logout

Students

Manage all student records

+ Add Student

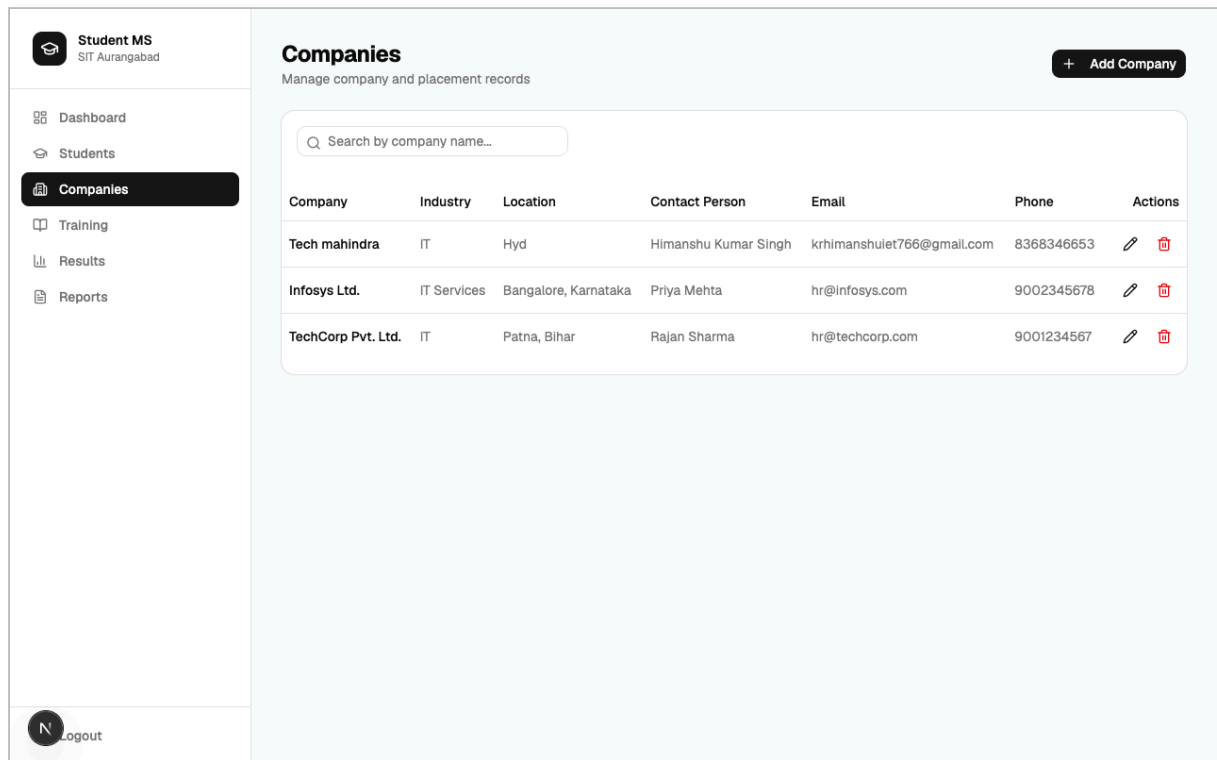
Q Search by name or roll no...

| Roll No. | Name                 | Email                      | Phone      | Course | Year   | Actions                                                                                                                                                                                                                                                     |
|----------|----------------------|----------------------------|------------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BCA012   | Himanshu Kumar Singh | krhimanshulet766@gmail.com | 8368346653 | BCA    | Year 3 |    |
| BCA004   | Om Prakash Kumar     | om@example.com             | 9876543213 | BCA    | Year 3 |    |
| BCA003   | Mohit Raj            | mohit@example.com          | 9876543212 | BCA    | Year 3 |    |
| BCA002   | Chirag Singh         | chirag@example.com         | 9876543211 | BCA    | Year 3 |    |
| BCA001   | Abhishek Kumar       | abhishek@example.com       | 9876543210 | BCA    | Year 3 |    |

*Figure 5.3: Student Module – Records Management*

### 5.2.4 Company Module

The company module stores all organisations where students are placed for training. Each record holds the company name, industry, location, contact person, email, and phone. Companies can be searched by name.



**Figure 5.4:** Company Module – Placement Records

### 5.2.5 Training Module

The training module tracks each student's internship. It records the student, company, project name, start/end dates, status (Ongoing/Completed), and a project description. Skeleton loaders appear during data fetch for better UX.



 **Student MS**  
SIT Aurangabad

Dashboard

Students

Companies

**Training**

Results

Reports

N

Logout

**Training**

Manage internship and training records

+ Add Training

Q Search by student name or roll no...

| Student                        | Company            | Project                     | Duration              | Status    | Actions                                                                                                                                                                 |
|--------------------------------|--------------------|-----------------------------|-----------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Himanshu Kumar Singh<br>BCA012 | Infosys Ltd.       | Python                      | 1/16/2026 – 7/16/2026 | Completed |   |
| Abhishek Kumar<br>BCA001       | TechCorp Pvt. Ltd. | Inventory Management System | 6/1/2025 – 8/31/2025  | Completed |   |
| Chirag Singh<br>BCA002         | Infosys Ltd.       | HR Portal                   | 7/1/2025 – 9/30/2025  | Ongoing   |   |

Figure 5.5: Training Module – Internship Tracking

### 5.2.6 Results Module

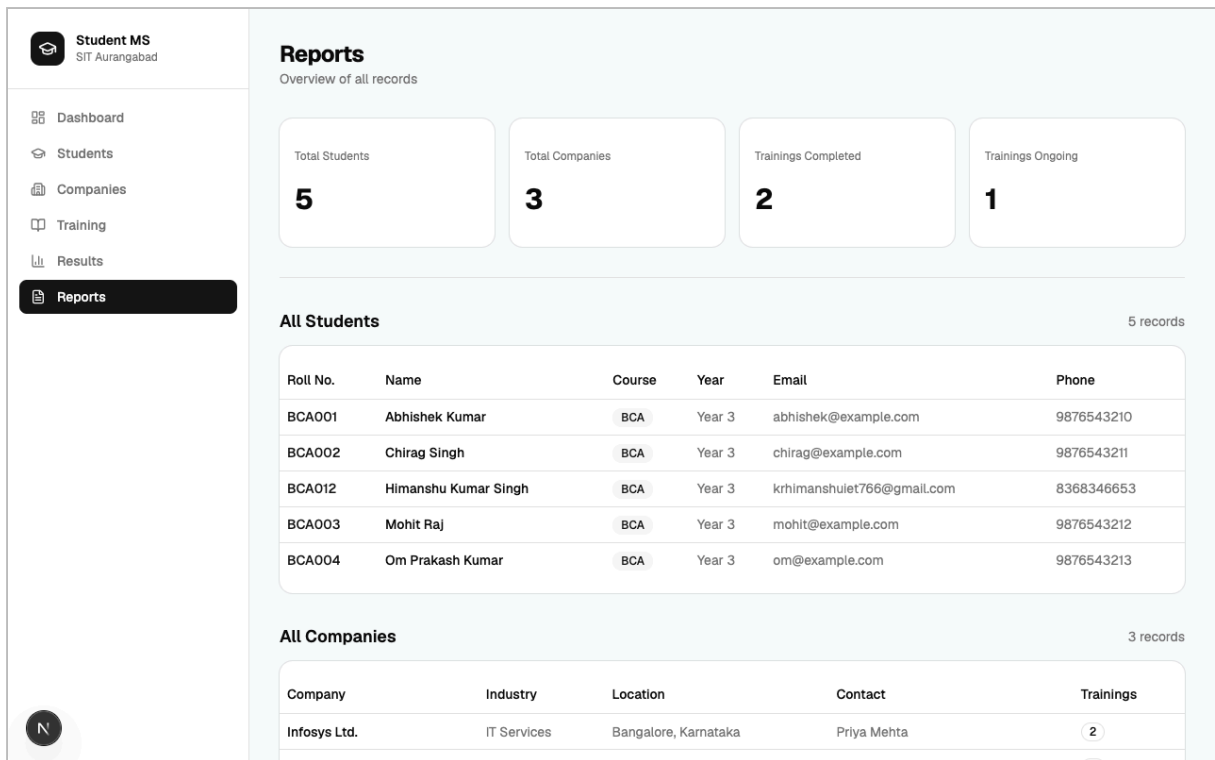
The results module records subject-wise academic performance. Each entry stores the student, subject, marks obtained, maximum marks, grade, semester, and year. Grades are displayed as colour-coded badges for quick visual scanning.

| Student                    | Subject        | Marks  | Grade | Semester | Year | Actions |
|----------------------------|----------------|--------|-------|----------|------|---------|
| Abhishek Kumar<br>BCA001   | Web Technology | 90/100 | A+    | Sem 5    | 2025 |         |
| Chirag Singh<br>BCA002     | DBMS           | 78/100 | B+    | Sem 5    | 2025 |         |
| Mohit Raj<br>BCA003        | DBMS           | 82/100 | A     | Sem 5    | 2025 |         |
| Om Prakash Kumar<br>BCA004 | Web Technology | 88/100 | A     | Sem 5    | 2025 |         |
| Abhishek Kumar<br>BCA001   | DBMS           | 85/100 | B+    | Sem 5    | 2025 |         |

*Figure 5.6: Results Module – Academic Performance*

### 5.2.7 Reports Module

The reports module presents a consolidated, read-only view of all data — students, companies, training records, and results — on a single page, giving administrators a complete institutional overview without switching modules.



*Figure 5.7: Reports Module – Consolidated View*

## Chapter 6

# Testing

---

### 6.1 Testing Strategy

The following testing approaches were adopted:

- **Unit Testing:** Individual API routes tested with direct HTTP requests for valid and invalid inputs.
- **Integration Testing:** Frontend form submissions and backend API interaction verified end-to-end.
- **User Acceptance Testing (UAT):** All modules manually tested by the project team against specified requirements.
- **Boundary Testing:** Empty fields, invalid types, and extreme values tested to confirm proper validation.
- **Security Testing:** Unauthenticated access to protected routes confirmed to redirect to login.

### 6.2 Test Cases

*Table 6.1: Test Cases – Authentication Module*

| TC#   | Test Case                       | Input            | Expected                  | Result |
|-------|---------------------------------|------------------|---------------------------|--------|
| TC-01 | Valid login                     | admin / admin123 | Redirect to dashboard     | ✓ Pass |
| TC-02 | Invalid password                | admin / wrong    | Error message shown       | ✓ Pass |
| TC-03 | Empty username                  | "" / admin123    | Form validation error     | ✓ Pass |
| TC-04 | Access /dashboard without login | No session       | Redirect to /login        | ✓ Pass |
| TC-05 | Logout                          | Click logout     | Session cleared, redirect | ✓ Pass |

*Table 6.2: Test Cases – Student Module*

| TC#   | Test Case                   | Input             | Expected                  | Result |
|-------|-----------------------------|-------------------|---------------------------|--------|
| TC-06 | Add student with valid data | All fields filled | Student record created    | ✓ Pass |
| TC-07 | Duplicate roll number       | Existing roll no. | Error: Roll number exists | ✓ Pass |
| TC-08 | Edit student details        | Modified fields   | Record updated in DB      | ✓ Pass |

| TC#   | Test Case      | Input          | Expected               | Result |
|-------|----------------|----------------|------------------------|--------|
| TC-09 | Delete student | Confirm delete | Record removed         | ✓ Pass |
| TC-10 | Search by name | Partial name   | Filtered results shown | ✓ Pass |

**Table 6.3:** *Test Cases – Company & Training Modules*

| TC#   | Module   | Test Case                        | Expected                      | Result |
|-------|----------|----------------------------------|-------------------------------|--------|
| TC-11 | Company  | Add company with valid data      | Company record created        | ✓ Pass |
| TC-12 | Company  | Duplicate company name           | Error: Company already exists | ✓ Pass |
| TC-13 | Company  | Edit company details             | Record updated                | ✓ Pass |
| TC-14 | Company  | Delete company                   | Company removed from list     | ✓ Pass |
| TC-15 | Training | Add training with valid data     | Training record created       | ✓ Pass |
| TC-16 | Training | Submit without selecting student | Validation error shown        | ✓ Pass |
| TC-17 | Training | Update status to Completed       | Badge shows Completed         | ✓ Pass |
| TC-18 | Training | Delete training record           | Record removed                | ✓ Pass |
| TC-19 | Results  | Add result with valid data       | Result record created         | ✓ Pass |
| TC-20 | Results  | Edit result grade                | Grade updated in list         | ✓ Pass |

### 6.2.1 Testing Summary

A total of 20 test cases were executed across Authentication, Student, Company, Training, and Results modules. All 20 test cases passed successfully, confirming that the system functions correctly as per the specified requirements. Error handling was verified with invalid and boundary inputs — in all cases the system responded with appropriate error messages rather than crashing or exposing internal details.

## Conclusion and Future Scope

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### 7.1 Conclusion

The **Student Training Management System** has been successfully designed, developed, and tested as part of the Bachelor of Computer Application final year project at Sityog Institute of Technology, Aurangabad. The system fulfils all the objectives outlined in Chapter 1 and effectively addresses the problems identified in the problem statement.

The project demonstrates the practical application of modern full-stack web development technologies — Next.js 16, TypeScript, Prisma ORM, Neon PostgreSQL, NextAuth v5, and shadcn/ui. The resulting application is a robust, secure, and user-friendly platform that enables administrators to manage student records, company placements, training assignments, and academic results with ease.

#### **Key achievements of this project include:**

- A fully functional web-based system accessible from any modern browser.
- Secure authentication with JWT sessions and bcrypt password hashing.
- Complete CRUD operations across all four data modules.
- Real-time search and filtering for quick data retrieval.
- Skeleton loading states for improved user experience.
- Side drawer forms for smooth, in-page data entry.
- Individual student profile pages with complete history.
- A consolidated reports page for institutional overview.
- Cloud-hosted PostgreSQL database for persistent, scalable data storage.

The system successfully replaces manual paper-based and spreadsheet-based record keeping, providing significant improvements in efficiency, accuracy, and accessibility of institutional data management.

## **7.2 Future Scope**

Several enhancements can be incorporated in future versions to make the system more comprehensive for larger institutional deployment:

- **Student Portal:** Allow students to log in and view their own training records, results, and profile.
- **Role-Based Access Control:** Multiple roles (Admin, Faculty, HOD) with different permission levels.
- **Email Notifications:** Automated emails to students when training or results are updated.
- **Attendance Management:** Daily attendance tracking with monthly summaries.
- **PDF Export:** Export individual student profiles, result sheets, and training certificates.
- **Mobile Application:** React Native companion app for on-the-go access.
- **Advanced Analytics:** Charts for placement trends, result distributions, and training statistics.
- **Multi-Campus Support:** Multiple departments and campuses within the same institution.
- **Placement Drive Module:** Manage drives, interview schedules, and offer letters.
- **SMS Notifications:** Alerts to students and parents about important events.



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